

Predicting the Next DineSafe Inspection Outcome from Inspection History: A Literature Review

Peeyusha Patki

B.Tech in Artificial Intelligence and Data Science
K J Somaiya Institute of Technology, Mumbai, India

1. Introduction

Toronto Public Health inspects food establishments in the city under the DineSafe program, and each inspection ends in a Pass, a Conditional Pass or a Closed notice [1]. The results are published as open data, together with a historical archive of earlier inspections [2]. Since the same establishment is inspected many times, the data form a sequence of inspections for each business. This project asks how much that inspection history helps to predict the outcome of an establishment's next inspection, and how the amount and recency of the history changes the result.

This review covers earlier work on predicting food inspection outcomes, studies that have used the Toronto DineSafe data, and methodological papers that shaped the design of the project. The aim is to place the project in context and to be clear about which parts of it are new and which are not.

2. Risk-Based Inspection and the Case for Prediction

Health departments have a limited number of inspectors, so they cannot give every establishment the same attention. Hartman [3] reviews how inspection agencies moved from simple risk categories toward models built on an establishment's inspection history, and later toward digital signals such as online reviews. A point that runs through this review is that such models are meant to support the scheduling of inspections, not to replace the judgement of inspectors. This framing is important for the present project, because a risk score is only useful if it helps decide where limited inspection time should go.

3. Forecasting Critical Violations in Chicago

The closest earlier work is the food inspection forecasting model developed by the City of Chicago with Allstate Insurance [4]. The model estimated which establishments were most likely to have a critical violation at their next routine inspection. Its inputs included past inspection results, the time since the previous inspection, business licence information, the inspector assigned, nearby sanitation complaints and burglaries, and recent weather. When the model was tested against the usual schedule, inspections ordered by the model found critical violations about a week earlier on average [3], [4]. The city released the code and data publicly.

Kannan, Shapiro and Bilgic [5] later carried out an independent analysis of the released Chicago model, looking at its data, predictor variables, performance measures and underlying assumptions. This work is relevant here for two reasons. First, it confirms that predicting a future inspection result from earlier information has already been done, so the idea on its own

is not new. Second, it shows why the evaluation must follow the order of time: a model that claims to predict the next inspection can only be given information that existed before that inspection took place.

4. Food Safety Compliance Studies in Other Settings

Oldroyd, Morris and Birkin studied food hygiene compliance in England and Wales. In their first study they used logistic regression and found that outlets in more deprived areas, takeaways and outlets in large urban areas were less likely to meet hygiene standards [6]. In a follow-up study they compared machine learning methods for predicting non-compliant outlets from openly available socio-demographic, business type and urbanness features [7]. These studies show that fixed characteristics of a business and its location carry useful signal, which is a helpful contrast to the history-based features used in this project.

More recent studies have used richer data and more complex models. Sari, Bader-El-Den and Ince [8] combined structured inspection records with the written deficiency notes from New York State inspections. They compared classical machine learning models, a BiLSTM and transformer models, and RoBERTa gave the highest F1 score of 0.96, with SHAP used to explain the predictions. Nalbone et al. [9] built a Bayesian network from official control records in southern Italy to predict which type of non-compliance was most likely to be found at a food establishment. Okumus, Mugaloglu and Kocak [10] modelled pass or fail outcomes of Chicago restaurant inspections using logistic regression and CatBoost, and interpreted the results with SHAP values. Together these studies show that machine learning, including interpretable gradient boosting, is already an established approach for food inspection outcomes.

5. Prediction from Other Data Sources

Sadilek et al. [11] followed a different route with FINDER, a system that used aggregated and anonymised web search and location data to identify restaurants that might be linked to foodborne illness. In field tests, restaurants flagged by FINDER were about 3.1 times as likely to be found unsafe as restaurants selected by existing methods. This work does not use inspection history, but it shows that a prediction is most valuable when it changes which establishments are inspected. It also points to a difference in what is being predicted. FINDER looks for signs of illness, whereas a model built on DineSafe records can only predict the recorded result of an inspection.

6. Earlier Studies of DineSafe

The DineSafe data have been studied before, but not for prediction. Besharah and Heacock [12] evaluated the DineSafe placard system and found no evidence that it had reduced the number of violations. They also reported that the share of establishments receiving a Conditional Pass twice within the study period rose from 59% to 68%. Thandi [13] compared restaurant infractions in Toronto and Vancouver using routine inspection reports and found significant differences between the two cities. Both studies are descriptive. In this review I did not find a published study that predicts future DineSafe outcomes from inspection history, although this does not rule out unpublished or very recent work.

7. Methodological Lessons for This Project

Temporal leakage. Kaufman et al. [14] describe leakage as information in the training data that would not be available at the time a prediction is made. With repeated inspections, a random split can let a model learn from an establishment's later inspections while it is tested on earlier ones. For this reason the project uses a strict time-ordered split. Each prediction uses only inspections that took place before the target inspection, and the model is trained on earlier periods and tested on later ones.

Class imbalance. Non-Pass results are a small share of inspections, about 7.5% of the test period in this project. Saito and Rehmsmeier [15] show that precision-recall analysis is more informative than ROC analysis when the positive class is rare. PR-AUC was therefore used as the main metric, with ROC-AUC reported alongside it.

Calibration. Van Calster et al. [16] argue that a model can rank cases well and still give misleading probabilities, so calibration should always be checked. This project checked calibration directly. Because the predicted probabilities were not well calibrated, the final model is presented as a tool for ranking inspections by risk and not as an estimate of the chance that an inspection will fail.

Models and interpretation. The project uses CatBoost [17], a gradient boosting method designed for data with categorical features, and SHAP values [18] to examine which features the final model relies on.

8. Research Gap and Contribution of This Project

The literature shows that predicting food inspection results with machine learning is not a new idea [4], [5], [7], [8], [10]. What has received less attention is how much of the predictive signal comes from an establishment's own inspection history, and whether a longer history adds anything beyond the most recent inspection. This is the question the project focuses on, using Toronto data.

To study this, the DineSafe export was rebuilt from 115,899 infraction-level rows into 77,475 inspections across 18,898 establishments, and the historical archive was linked through the older establishment identifier. Models given different amounts of history were then compared under the same time-ordered split. Using only the previous inspection gave a test PR-AUC of 0.127. Adding 365 days of history raised it to 0.162, and using 730 days raised it to 0.173. The tuned CatBoost model reached a PR-AUC of 0.174 and a ROC-AUC of 0.734, and the 10% of inspections it ranked as highest risk contained about 28% of all non-pass inspections. These results suggest that inspection history carries useful information, with most of the gain coming from the first year of history and a smaller gain from the second. Table 1 summarises the studies discussed in this review.

Table 1. Summary of the main studies reviewed

Study	Setting and data	Target	Method	Relevance to this project
City of Chicago [4]	Chicago routine food inspections	Critical violation at next inspection	Regularised regression model	Closest earlier example of next-inspection

Study	Setting and data	Target	Method	Relevance to this project
				prediction
Kannan et al. [5]	Released Chicago model and data	Critical violation	Independent analysis	Shows why evaluation must follow time order
Oldroyd et al. [6]	England and Wales hygiene ratings	Non-compliant outlet	Logistic regression	Location and business type carry signal
Oldroyd et al. [7]	England and Wales hygiene ratings	Non-compliant outlet	Comparison of machine learning methods	Compliance prediction with a rare positive class
Sari et al. [8]	New York State inspections	Inspection grade	LightGBM, BiLSTM, transformers, SHAP	Recent approach using inspection text
Nalbone et al. [9]	Official controls, southern Italy	Type of non-compliance	Bayesian network	Predicts the kind of problem, not only pass or fail
Okumus et al. [10]	Chicago restaurant inspections	Pass or fail	Logistic regression, CatBoost, SHAP	Recent interpretable model of inspection outcomes
Sadilek et al. [11]	Web search and location data, USA	Unsafe restaurant	Machine-learned screening	Prediction used to decide which places to inspect
Besharah and Heacock [12]	Toronto DineSafe	Repeat Conditional Pass	Descriptive comparison	Earlier DineSafe study, not predictive
Thandi [13]	Toronto and Vancouver reports	Infraction counts	Statistical comparison	Earlier DineSafe study, not predictive

9. Limitations of This Review

This was a targeted review rather than a formal systematic review. The search focused on English language sources that were directly related to food inspection prediction and to the DineSafe data, so relevant work in other languages or in grey literature may have been missed. Several of the studies cited were published in 2026, which shows that the field is moving quickly, and newer work may appear after this review was written.

10. Conclusion

Earlier research has already shown that food inspection outcomes can be predicted, that inspection history is a useful input, and that predictions can help target inspections. The contribution of this project is therefore not the prediction task itself. It is a careful Toronto case study that measures how much an establishment's inspection history adds, how that depends on the length of history used, and how useful the resulting risk ranking is under a strict time-ordered evaluation, while being clear that the model predicts a recorded inspection result and not the true level of food safety.

References

- [1] City of Toronto, "DineSafe," Toronto Public Health. [Online]. Available: <https://www.toronto.ca/community-people/health-wellness-care/health-programs-advice/food-safety/dinesafe/> (accessed Sep. 16, 2026).
- [2] City of Toronto, "DineSafe," City of Toronto Open Data Portal. [Online]. Available: <https://open.toronto.ca/dataset/dinesafe/> (accessed Sep. 16, 2026).
- [3] J. Hartman, "Predictive models for food code violations," *Food Protection Trends*, vol. 40, no. 1, pp. 56-65, Jan./Feb. 2020. [Online]. Available: <https://www.foodprotection.org/files/food-protection-trends/jan-feb-20-hartman.pdf>
- [4] T. Schenk Jr., G. Leynes, A. Solanki, S. Collins, G. Smart, B. Albright, and D. Crippin, "Forecasting restaurants with critical violations in Chicago," City of Chicago and Allstate Insurance, Chicago, IL, USA, Tech. Rep. [Online]. Available: <https://github.com/Chicago/food-inspections-evaluation> (accessed Sep. 16, 2026).
- [5] V. Kannan, M. A. Shapiro, and M. Bilgic, "Hindsight analysis of the Chicago food inspection forecasting model," arXiv:1910.04906, 2019. [Online]. Available: <https://arxiv.org/abs/1910.04906>
- [6] R. A. Oldroyd, M. A. Morris, and M. Birkin, "Food safety vulnerability: Neighbourhood determinants of non-compliant establishments in England and Wales," *Health & Place*, vol. 63, Art. no. 102325, 2020, doi: 10.1016/j.healthplace.2020.102325. [Online]. Available: <https://doi.org/10.1016/j.healthplace.2020.102325>
- [7] R. A. Oldroyd, M. A. Morris, and M. Birkin, "Predicting food safety compliance for informed food outlet inspections: A machine learning approach," *Int. J. Environ. Res. Public Health*, vol. 18, no. 23, Art. no. 12635, 2021, doi: 10.3390/ijerph182312635. [Online]. Available: <https://doi.org/10.3390/ijerph182312635>
- [8] O. F. Sari, M. Bader-El-Den, and V. Ince, "Explainable transformer-based modelling for pathogen-oriented food safety inspection grade prediction using New York State open data," *Foods*, vol. 15, no. 2, Art. no. 223, Jan. 2026, doi: 10.3390/foods15020223. [Online]. Available: <https://doi.org/10.3390/foods15020223>
- [9] L. Nalbone et al., "Machine learning and food inspection: Use of Bayesian Network modeling to support official controls in the food industries," *Italian Journal of Food Safety*, 2026. [Online]. Available: <https://www.pagepressjournals.org/ijfs/article/view/13491>
- [10] B. Okumus, E. Mugaloglu, and E. Kocak, "Pass or fail? Modeling restaurant inspection outcomes through interpretable machine learning," *Int. J. Hospitality Management*, vol. 140, Art. no. 104810, 2027, doi: 10.1016/j.ijhm.2026.104810. [Online]. Available: <https://doi.org/10.1016/j.ijhm.2026.104810>
- [11] A. Sadilek et al., "Machine-learned epidemiology: Real-time detection of foodborne illness at scale," *npj Digital Medicine*, vol. 1, Art. no. 36, 2018, doi: 10.1038/s41746-018-0045-1. [Online]. Available: <https://doi.org/10.1038/s41746-018-0045-1>
- [12] A. Besharah and H. Heacock, "Dinesafe Toronto: An evaluation of the placard system," *BCIT Environmental Public Health Journal*, 2015, doi: 10.47339/ephj.2015.117. [Online]. Available: <https://journals.bcit.ca/index.php/ehj/article/view/117>
- [13] A. Thandi, "A statistical comparison of restaurant infractions between Toronto and Vancouver," *BCIT Environmental Public Health Journal*, 2021. [Online]. Available: <https://journals.bcit.ca/index.php/ehj/article/view/193>
- [14] S. Kaufman, S. Rosset, C. Perlich, and O. Stitelman, "Leakage in data mining: Formulation, detection, and avoidance," *ACM Trans. Knowl. Discov. Data*, vol. 6, no. 4, Art. no. 15, 2012, doi: 10.1145/2382577.2382579. [Online]. Available: <https://doi.org/10.1145/2382577.2382579>
- [15] T. Saito and M. Rehmsmeier, "The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets," *PLoS ONE*, vol. 10, no. 3, Art. no.

- e0118432, 2015, doi: 10.1371/journal.pone.0118432. [Online]. Available: <https://doi.org/10.1371/journal.pone.0118432>
- [16] B. Van Calster, D. J. McLernon, M. van Smeden, L. Wynants, and E. W. Steyerberg, "Calibration: The Achilles heel of predictive analytics," *BMC Medicine*, vol. 17, Art. no. 230, 2019, doi: 10.1186/s12916-019-1466-7. [Online]. Available: <https://doi.org/10.1186/s12916-019-1466-7>
 - [17] L. Prokhorenkova, G. Gusev, A. Vorobev, A. V. Dorogush, and A. Gulin, "CatBoost: Unbiased boosting with categorical features," in *Proc. Advances in Neural Information Processing Systems (NeurIPS)*, 2018. [Online]. Available: <https://arxiv.org/abs/1706.09516>
 - [18] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in *Proc. Advances in Neural Information Processing Systems (NIPS)*, 2017. [Online]. Available: <https://arxiv.org/abs/1705.07874>